

Generative AI in Computer Vision

- What is Generative AI in computer vision
- Synthetic Data Generation
 - Data Augmentation with Image Processing
 - Synthetic data Generation with AI algorithm
 - Auto-Encoder
 - Different type of Autoencoders
 - Generative Adversarial Network(GAN)

What is Generative AI?

- A branch of artificial intelligence focused on creating (or “generating”) new data—images, videos, or 3D objects—that resemble real-world examples.

Key Techniques

- **Data Augmentation**
- **Variational Autoencoders (VAEs):** Learn a latent representation of data and can generate new samples by sampling from this latent space.
- **Generative Adversarial Networks (GANs):** Pit a “generator” against a “discriminator,” leading to highly realistic images and deepfakes.
- **Diffusion Models:** Iteratively refine noise into a coherent image, often producing high-fidelity, diverse outputs.

- **Image Generation & Editing:** From creating photorealistic portraits and landscapes to changing attributes (e.g., hair color, lighting).
- **Data Augmentation:** Synthesizing more training data to improve model performance, especially when real data is scarce.
- **Super-Resolution & Inpainting:** Enhancing image quality or filling missing regions seamlessly.
- **Style Transfer:** Combining the visual style of one image (e.g., Van Gogh) with the content of another (e.g., a photo).

1. **Expanded Creative Potential**

- Enables artists, designers, and filmmakers to rapidly prototype or generate new content without starting from scratch.
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2. **Enhanced Model Training**

- Synthetic data helps when real-world data is scarce or expensive. Boosts performance by providing diverse, controlled training samples.

What is Data Augmentation?

Definition: Data augmentation is the process of artificially expanding your training dataset by creating modified versions of existing samples.

Purpose:

- Increase dataset size when collecting new data is expensive or infeasible.
- Improve model generalization and reduce overfitting.

Common Techniques:

1. **Geometric Transformations:** Rotation, flipping, cropping, translation, scaling.
2. **Color & Lighting Adjustments:** Brightness, contrast, saturation, hue shifts.
3. **Noise Injection:** Adding Gaussian noise, random occlusions, or blur.
4. **Advanced Methods:**
 - **CutMix / Mixup:** Combining parts of different images/labels.

Type of Data Augmentation Methods

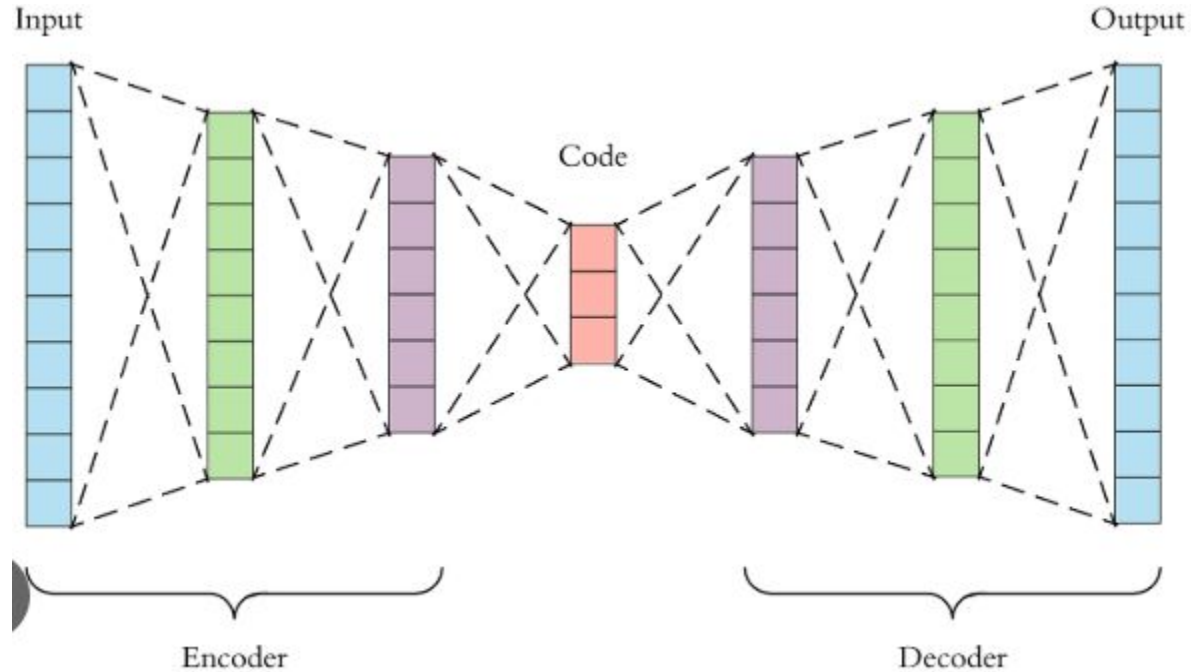
Image Augmentation we use in our class:

- Rotation
 - Flipping
 - Random crops,
 - Color jitter, etc.
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- Widely used in computer vision tasks (classification, detection, segmentation)

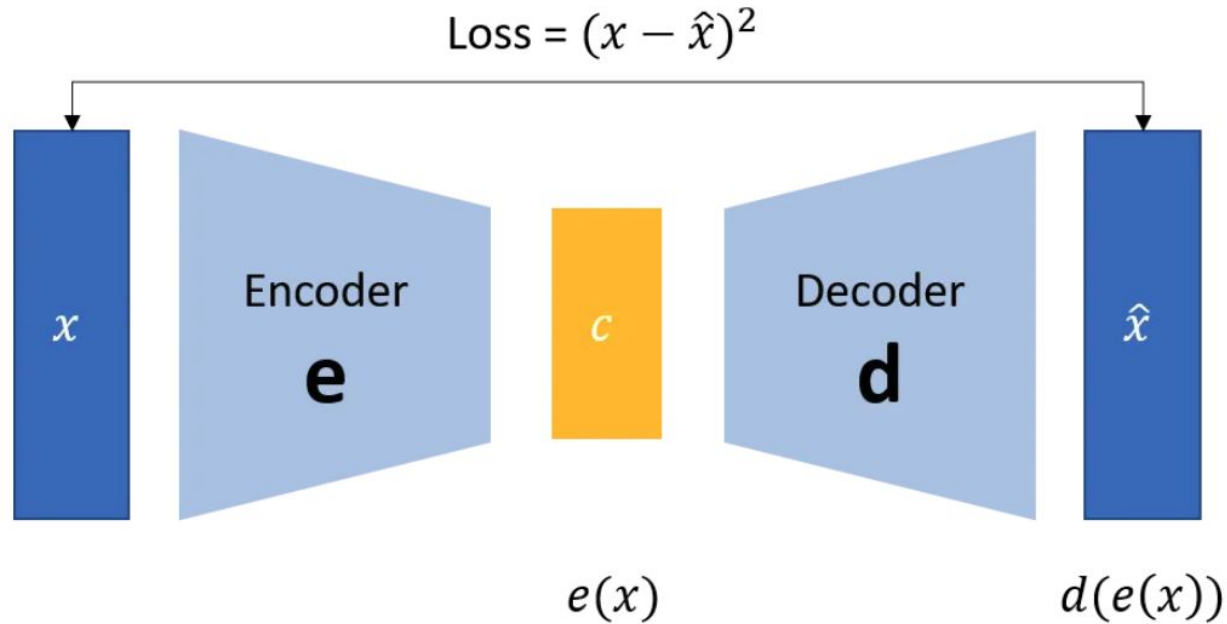
1- Basic Autoencoder

- **Idea:** Learn to compress (encode) input data into a lower-dimensional latent representation and then reconstruct (decode) it back to the original dimension.
- **Architecture:**
 - An **encoder** network that reduces input dimensionality.
 - A **decoder** network that reconstructs the original input from the reduced representation.

Basic Auto Encoder



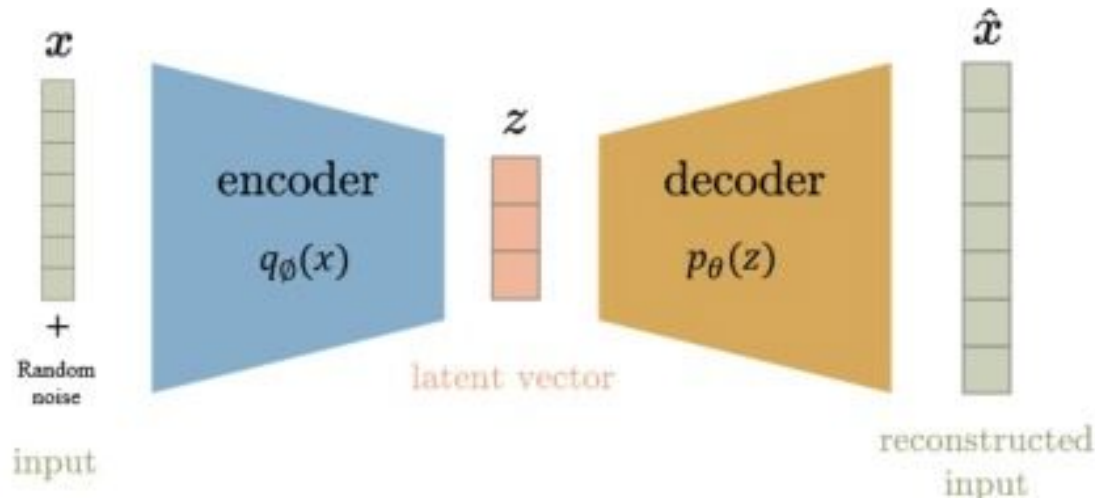
Basic Auto-Encoder



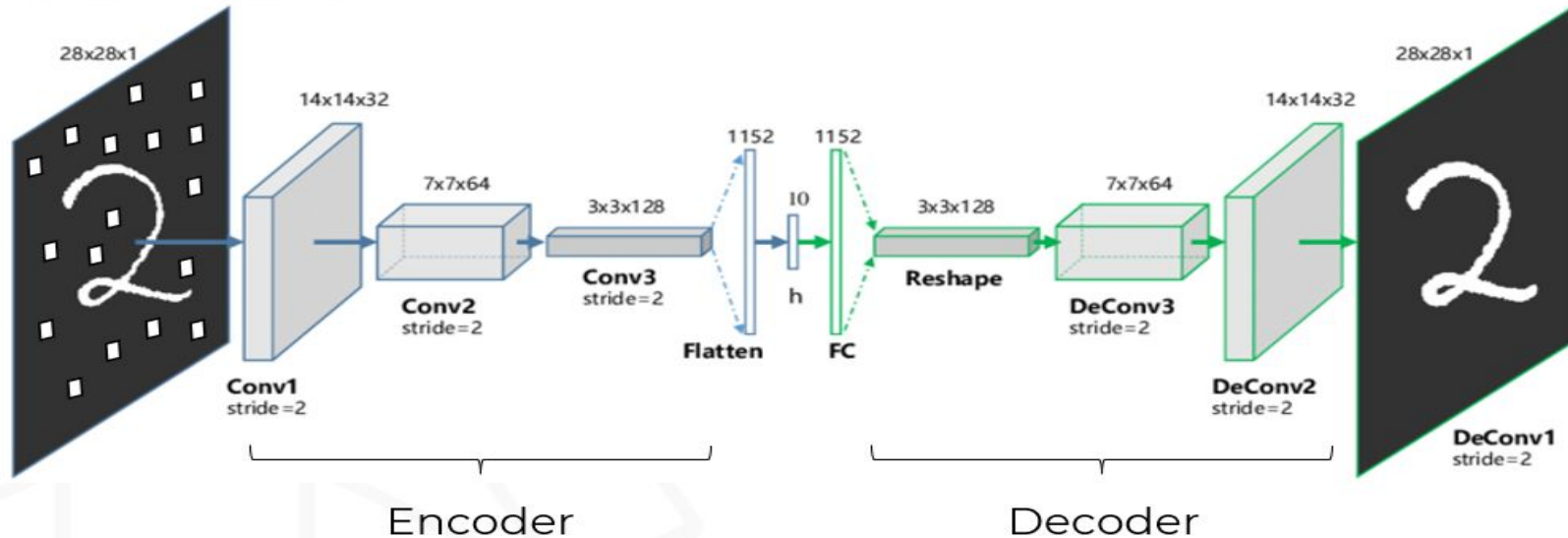
2. Denoising Autoencoder (DAE)

- **Idea:** Train the autoencoder to remove noise from corrupted inputs. The goal is to learn robust features.
- **Key Difference:** The input given to the network is a *noisy or corrupted* version of the data, but the target is the *clean* version.
- **Benefit:** Improves generalization and robustness to noise. It also helps the autoencoder avoid simply learning an identity function.

Denoising AUto-Encoder



Denoising Autoencoders

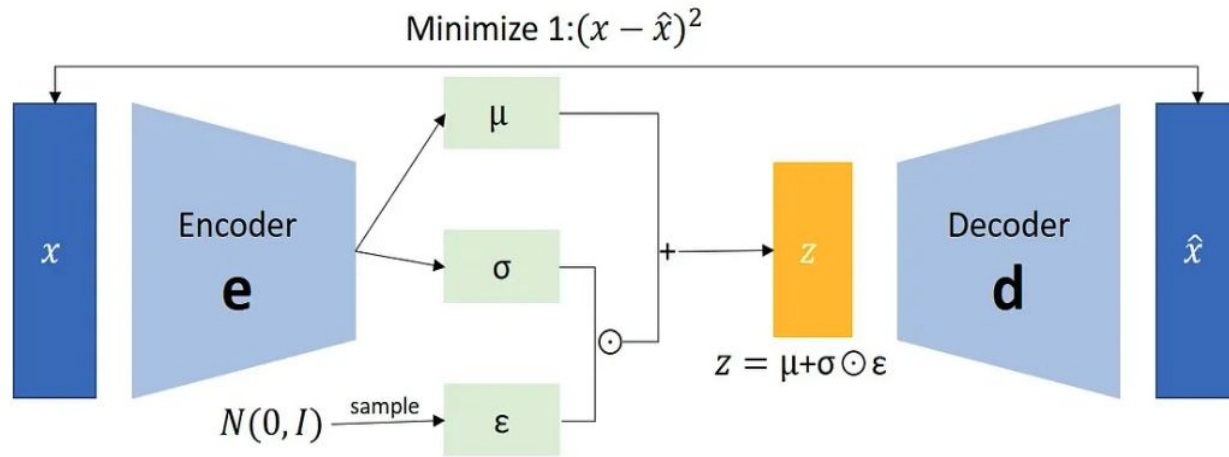


Different Type of Auto-Encoders:

3- Variational Autoencoder (VAE)

- **Idea:** Learn the *distribution* of the input data by enforcing a probabilistic latent space. Instead of encoding an input into a single latent vector, the VAE encodes it into a *distribution* (mean and variance).
- **Key Difference:** Uses a **reparameterization trick** to learn continuous latent variables, allowing the model to sample from the latent distribution and generate new data.
- **Benefit:** Very popular for *generative modeling*. It can generate novel, meaningful samples (e.g., new images or text).

Variational Auto-Encoder



Minimize 2: $\frac{1}{2} \sum_{i=1}^N (\exp(\sigma_i) - (1 + \sigma_i) + \mu_i^2)$

Different Type of Auto-Encoders:

4-Adversarial Autoencoder (AAE)

- **Idea:** Combine the autoencoder framework with a *Generative Adversarial Network (GAN)*. The autoencoder part reconstructs inputs, while a discriminator tries to distinguish latent encodings from real samples from some prior distribution.
- **Key Difference:** An adversarial loss encourages the latent space to match a given prior distribution (e.g., Gaussian).
- **Benefit:** Produces flexible latent representations that still allow for generative capabilities (similar to VAEs, but with a GAN-style approach to match distributions).

Questions?